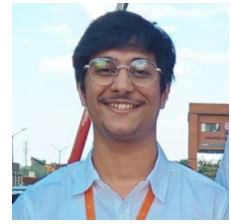


SHASHWAT SOLANKI

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Full-stack developer with hands-on ISRO R&D experience building performance-critical geospatial systems, real-time 3D visualization tools, and data-intensive backend services for satellite and urban analysis.



EDUCATION

PES University

B.Tech in Computer Science & Engineering

CGPA: 7.97

Bengaluru, India

Aug 2023 – Present

EXPERIENCE — SPACE APPLICATIONS CENTRE (ISRO)

Research Intern

Jul 2025 – Aug 2025

Geospatial Systems & Visualization

Ahmedabad, India

- Developed **Vedant**, a CesiumJS-based geospatial analysis platform integrating Terrain Profile, Viewshed Analysis, Solar Rooftop Potential, and plugin-driven architecture.
- Engineered terrain profiling algorithms to compute elevation cross-sections for remote sensing workflows.
- Implemented viewshed analysis for line-of-sight computation supporting satellite communication and urban visibility use cases.
- Designed a solar rooftop potential estimator using sun-angle modeling and 3D building datasets.
- Built **RIDaM (Raster Imagery Data Management)**, a full-stack system for managing large-scale raster satellite datasets and metadata pipelines.

PROJECTS

Vedant — Geospatial Analysis Tool (ISRO) | *CesiumJS, Vue.js, ElectronJS, RxJS*

- Integrated multiple geospatial analysis modules into a unified 3D simulation environment.
- Optimized interactive 3D rendering for large raster datasets, reducing frame drops during intensive visual analysis.
- Worked in a 2-member team, owning frontend architecture and CesiumJS integration.

RIDaM — Raster Imagery Data Management (ISRO) | *Node.js, PostgreSQL, Vue.js*

- Built a backend-driven platform for parsing, storing, and managing multi-source raster satellite imagery.
- Designed optimized PostgreSQL schemas and indexing strategies for imagery metadata and spatial attributes.

Console-based E-Commerce Platform | *C, Data Structures*

GitHub

- Built a modular e-commerce system in C with product catalog, cart handling, and order processing.
- Implemented file-based persistence and a basic recommendation mechanism using purchase history.

ACHIEVEMENTS

- Secured **4th place** in a competitive programming contest (CodeChef) at PES University among 60+ participants.
- Secured **6th place** in Ather Hackathon by developing a mountable sensor device that differentiates walking vs driving, tracking real-time speed, distance, angle, and GPS location.

CERTIFICATIONS

- Geospatial Data Analysis Internship — **ISRO**
- Complete Python Bootcamp — Udemy
- C Programming Bootcamp — Udemy
- Problem Solving (Basic) — HackerRank

SKILLS

Geospatial & Visualization: CesiumJS, ElectronJS

Backend & Systems: Node.js, Express.js, PostgreSQL

Frontend: Vue.js, React.js, HTML, CSS

Database: PostgreSQL (schema design, indexing), SQL, MongoDB

Languages: C, C++, Python, JavaScript

Tools: Git, GitHub, GitHub Actions (CI/CD), VS Code